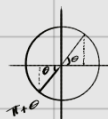


$$1) b) \int_{-1}^1 \int_{-1}^1 \int_{-1}^1 x^2 y^2 z^2 dx dy dz = \int_{-1}^1 \int_{-1}^1 y^2 z^2 \left(\frac{x^3}{3} \Big|_{-1}^1 \right) dy dz = \int_{-1}^1 \int_{-1}^1 y^2 z^2 \left(\frac{y^3}{3} \Big|_{-1}^1 \right) dz = \int_{-1}^1 y^2 dz = \frac{y}{3} \Big|_{-1}^1 = \frac{2}{3}$$



$$c) \int_0^\pi \int_0^\pi \int_0^\pi \sin(x+y+z) dx dy dz = \int_0^\pi \int_0^\pi [-\cos(x+y+z)]_0^\pi dy dz = \int_0^\pi \int_0^\pi z \cos(y+z) dy dz = \int_0^\pi [z \sin(y+z)]_0^\pi dz = \int_0^\pi -z \sin(z) dz = -8$$

$$\textcircled{I} -\cos(x+y+z) \Big|_0^\pi = (-\cos(\pi+y+z) - (-\cos(0+y+z))) = \underbrace{\cos(y+z)}_{\cos \theta} - \underbrace{\cos(\pi+y+z)}_{-\cos \theta} = z \cos \theta = z \cos(y+z)$$



$$\theta = y+z$$

$$\cos(\pi+\theta) = -\cos \theta$$



$$\textcircled{II} [\sin(y+z)]_0^\pi = \sin(\pi+z) - \sin(z) = -2 \sin z$$



$$\rightarrow \sin(\beta) = -\sin(\pi+\beta) \rightarrow \sin(\pi+\beta) = -\sin \beta$$

$$\textcircled{III} -4 \left(-\cos z \Big|_0^\pi \right) = -4 \left(-\cos \pi - (-\cos 0) \right) = -8$$

